ELEC 49X Project Outline

Section 1) This section is to be filled by the students:
Project title: AI for a Weight-Lifting Athlete
Group #: 53

Group members:
1- Rhys Blakeney
2- Luke Hunker
3- Avedis Boudakian
4- Ajmal Niazi

1. Problem hypothesis

RPE is a crucial metric to determine an athlete's training intensity, fatigue, and progression. The scale is highly subjective, making it difficult to assess. Using an AI-based solution to accurately calculate RPE will optimize professional and beginner athletes' workout routines and training, so they achieve the best results.

2. Solution hypothesis

The solution to this project is to create a computer vision algorithm that will take video input of the user squatting, bench pressing, or deadlift and will output the RPE of the set. Using bar speed, facial expressions, rep quality, tempo change, and breathing, the algorithm will calculate the RPE.

3. Components and Budget

To train the machine learning model for this project, we may require cloud GPU resources. We will also need a training dataset, which we plan to create ourselves or source from freely available information online. Overall, the budget is expected to be below the allotted $600.

Section 2) This section is to be filled by the supervisor:
Supervisor Agreement:

I, Prof. __Karen Rudie_____ , have agreed to supervise this team/project.

Supervisor Signature: _____

Date: __September 20, 2025_____